Power BI Dashboard Portfolio

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Regional share change in energy drink consumption

1.1 Visualisation 1

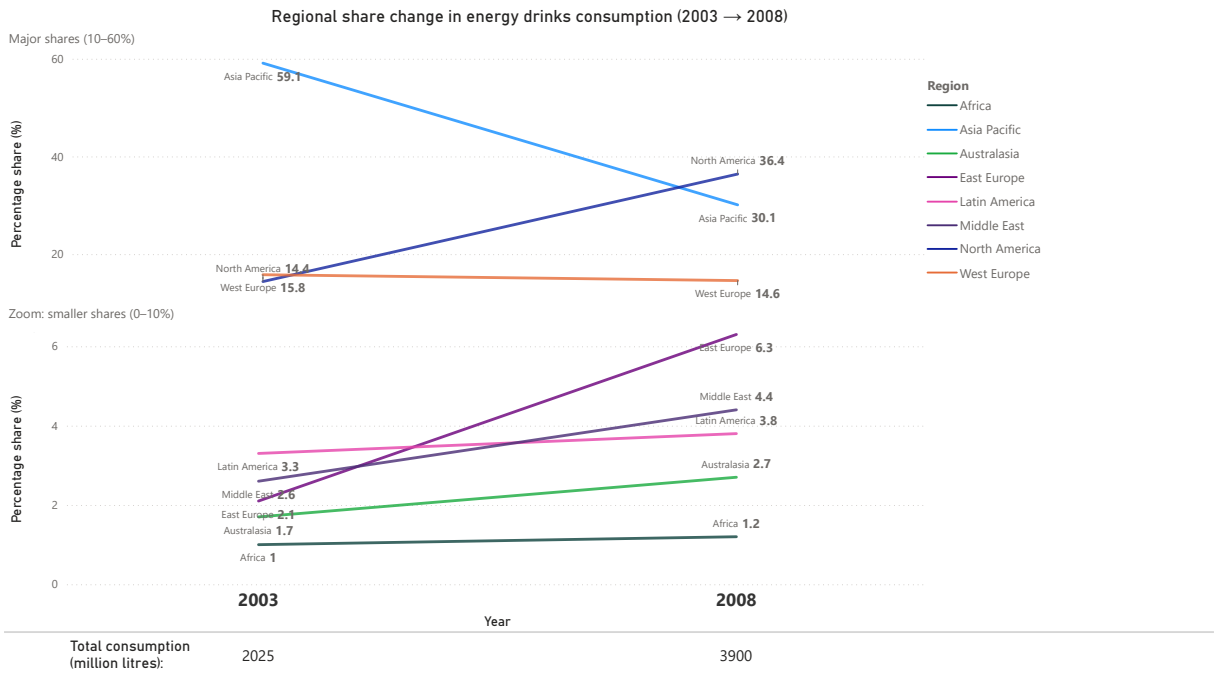

Figure 1: Regional share change in energy drink consumption (2003–2008).

1.2 Report 1

The data in Figure 1 compares regional percentage shares of total energy drink consumption in 2003 and 2008 and reports the overall total volume (million litres) for each year. Regional shares shifted over the period: some regions lost share (e.g., Asia Pacific and Western Europe), while others gained share (e.g., North America and Eastern Europe). The total volume also increased substantially, as indicated by the total litres KPI.

To represent change more effectively, I used a slope chart, where each region is shown as a coloured line connecting its percentage share in 2003 and 2008 (a five-year change between two time points). The visual is divided into two panels to improve readability: the lower panel zooms into 0–10% to make small regional shares visible, while the upper panel shows the larger shares (10–60%) on the original scale, as indicated by the axis.

The total volume is presented as a small KPI below the chart (under the year labels). Placing it separately improves readability and avoids mixing different units (percentages vs litres) on the same axis. The lines are also labelled with region names and percentage values to reduce reliance on colour alone. While labels can add some visual noise, they improve accessibility (e.g., for viewers with colour-vision deficiencies) and make small changes easier to interpret. This visualisation is useful for business and market analysis because it summarises the global distribution of consumption share and how it changed between 2003 and 2008.

This redesign communicates change between the two years more clearly than the original. It is easier to compare the same region across time because the two values are aligned and connected. In contrast, the original used two pie charts, which are difficult to compare across years: viewers must judge angles and areas, and small differences are hard to detect. With multiple categories, pie charts can become overwhelming and do not support precise comparisons at a glance. The original title ("Global energy drinks by years 2003–08") was also misleading because it did not clearly indicate that the chart shows changes in regional shares. I

renamed it to “Regional share change in energy drink consumption (2003 → 2008)” to state the purpose and the comparison explicitly.

Pre-attentive attributes were used to guide attention: the two years are bolded to highlight the comparison points, and key values are placed near the endpoints for quick reading. Smaller fonts were used for secondary information (e.g., the litres KPI and some labels) to reduce clutter. Consistent colour assignment across both panels supports the Gestalt principle of similarity, while connecting points with lines supports continuity and helps the viewer track each region from 2003 to 2008.

Top 5 shopping malls in Istanbul by monthly revenue

2.1 Visualisation 2

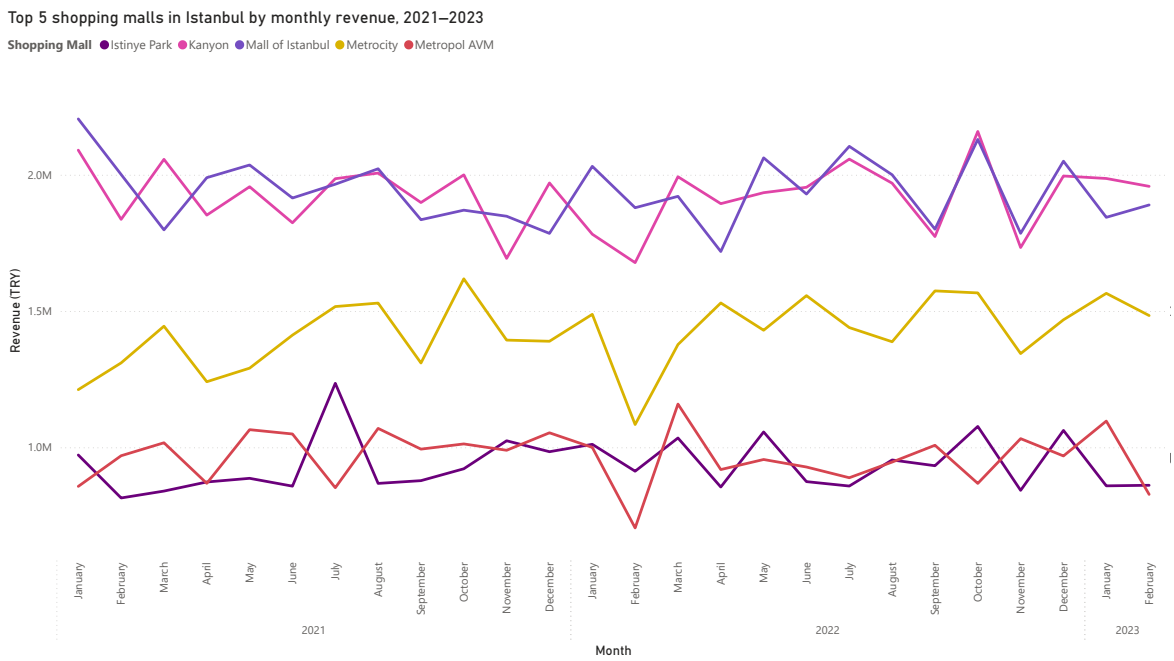

Figure 2: Top 5 shopping malls in Istanbul by monthly revenue (2021–2023).

Link to the visualisation: Public (Noroff) OneDrive folder link

2.2 Report 2

From the Customer Shopping Dataset – Retail Sales Data, a static line chart on Figure 2 was created to illustrate monthly revenue trends in Istanbul from 2021 to 2023 for the five highest-revenue shopping malls. The intended audience is mall managers and business stakeholders who need a clear overview of sales performance over time. The main purpose of the visualisation is to identify when revenue increases or decreases across the year and to compare which malls generate the highest monthly revenue.

Revenue was derived to better represent the value of sales per transaction. A calculated field was created as Revenue = price × quantity (TRY), which captures the total sales value rather than unit price alone. The time component was incorporated by aggregating invoice dates to monthly granularity. This monthly aggregation reduces short-term noise and makes longer-term patterns easier to interpret. To improve readability and avoid visual clutter, the data were filtered to the top five malls by total revenue over the full period, enabling stakeholders to focus on the largest contributors.

A line chart was selected because it is well suited for time-series data and supports comparison of month-to-month fluctuations. Different colours represent different shopping malls and a legend is provided for identification, allowing the reader to trace each mall's trend across the timeline. Exact data labels were not added to every point to keep the chart uncluttered; instead, the y-axis communicates the magnitude of revenue in Turkish Lira (TRY), while the overall trend and relative differences remain visually clear. March 2023 was excluded because it represents an incomplete month in the dataset and would bias the month-to-month comparison.

This visualisation answers three key questions: which malls consistently lead in monthly revenue, when major peaks or dips occur, and whether multiple malls show similar movements that could reflect broader seasonal or external effects. For example, the chart highlights a pronounced peak for Istinye Park in July 2021 and a

clear drop across malls around February 2022, which can be further investigated against promotions, holidays, or external events.

Limitations include the exclusion of the remaining five malls, which means the visual does not represent the full market distribution. In addition, monthly aggregation can obscure weekly or daily seasonality and does not explain the underlying causes of the observed changes.

From the design principles covered in the course, the visualisation applies continuity and connectedness by using lines to show how revenue changes from month to month. A consistent time scale and continuous line paths support pattern detection (peaks, dips, and trends) across the full period. Colour is used to differentiate the five malls and is paired with a clear legend to aid identification. To reduce clutter and cognitive load, only the top five malls are shown and point labels are omitted, keeping the focus on temporal patterns rather than exact values at every month.

Data Cleaning Reports

3.1 Data Cleaning Report 1

Data preparation was performed in Power BI using Power Query. After importing the text file, the first row was promoted to headers and the comma-separated values were parsed into three fields: Region, Percentage, and Year. Data types were corrected by setting Percentage to a decimal number and converting the date field to extract the year as an integer. The 'Total Litres' entry was treated as a separate metric (volume rather than percentage). To avoid mixing incompatible units in the same column, rows where Region = 'Total Litres' were filtered into a separate table containing Year and Total litres. This totals table was then merged back into the regional share table using a left outer join on Year, adding Total litres as a contextual column (repeated for each region within a year). No values were rescaled; the original ratios and totals were preserved.

3.2 Data Cleaning Reports 2

All data preparation and visualisation were performed in Power BI. The `invoice_date` field was converted to the `Date` data type using *Using Locale* because inconsistent date formatting prevented Power BI from reliably parsing some records. The numeric fields `price` and `quantity` were converted from text to appropriate numeric types. A derived variable, `Revenue`, was created to represent transaction value, calculated as `Revenue = price` $\times$ `quantity` (TRY). Finally, transactions were aggregated to monthly totals to support time-series trend analysis.